

COMPUTER PROGRAMMING

LAB NO: 06

LAB TITLE

Implementation of For Loop and Nested For Loop in C++

1.OBJECTIVE

The main objective of this lab is to understand how loops work in C++. Students learn how to use the for loop to execute a block of code multiple times. Another important objective is to understand the concept of **nested loops**, where one loop runs inside another loop. This lab also helps in developing logical thinking by solving problems like number printing, prime number checking, and pattern generation.

2. THEORY

A **for loop** is a control structure used in programming when the number of iterations is known. It consists of three parts: initialization, condition, and update. The loop starts with initialization, then checks the condition, and if the condition is true, it executes the body. After execution, the update statement runs, and the process repeats until the condition becomes false.

A **nested for loop** means placing one for loop inside another. The inner loop runs completely for each iteration of the outer loop. Nested loops are commonly used in problems like patterns, tables, and matrices. These loops help in solving complex problems using simple logic.

The For Loop:

1. A for loop is a repetition control structure that allows you to efficiently write a loop that needs to execute a specific number of times
2. The for loop is used when a value runs from a starting point to an ending point with a

constant increment or decrement.

Syntax:

```
for (initialization; condition; update variable)
{
    Statements
}
```

The Nested for loop:

A nested for loop means a for loop inside another for loop.

Syntax:

```
for (initialization; condition; increment) {
    for (initialization; condition; increment) {
        // inner loop code
    }
}
```

3. SOFTWARE

- 1.DEV-C++ IDE
- 2.C++ Compiler

4. PROCEDURE / PROCESS

Display Numbers Using For Loop:

- 1.Write a C++ program that uses a while loop to display numbers from 1 to 10.
2. Display each number on a separate line

```
.
#include <iostream> // input/output library
using namespace std;
int main() {
    int i = 1; // counter variable ko 1 se start kar rahe hain
    while (i <= 10) { // jab tak i 10 se chota ya barabar hai loop chalega
        cout << i << endl; // current number display karna
        i++; // i ko 1 se increase karna (next number ke liye)
    }
}
```

```

    return 0; // program end
}

```

Nested for Loop (Multiplication Table):

1. Write a C++ program that generates a multiplication table from 1 to 10.
2. Use nested for loops to iterate through the rows and columns of the table.
3. Display the multiplication table in a formatted manner.

```

#include <iostream>
using namespace std;
int main() {
    for(int i = 1; i <= 10; i++) { // outer loop: tables 1 to 10
        cout << "Table of " << i << endl; // table heading
        for(int j = 1; j <= 10; j++) { // inner loop: 1 to 10 multiplication
            cout << i << " x " << j << " = " << i * j << endl; // multiplication result
        }
        cout << endl; // next table spacing
    }
    return 0; // program end
}

```

Sum Using While Loop:

1. Write a C++ program that prompts the user to enter a positive integer.
2. Use a for loop to calculate the sum of all numbers from 1 to the entered integer.
3. Display the calculated sum.

```

#include <iostream> // input/output library
using namespace std;
int main() {
    int n; // user input number
    int sum = 0; // sum store karne ke liye variable
    cout << "Enter a positive integer: "; // user se input mangna
    cin >> n; // input lena
    for(int i = 1; i <= n; i++) { // 1 se n tak loop chalana
        sum = sum + i; // numbers ko add karna
    }
    cout << "Sum is: " << sum << endl; // final result display karna
}

```

```
return 0; // program end
}
```

Nested Pattern Printing Using Nested For Loop:

1. Write a C++ program that uses nested for loops to print a pattern of asterisks.
2. The pattern should form a triangle shape, with each row having an increasing number
3. The number of rows in the pattern should be determined by user input.

```
#include <iostream> // input/output library
using namespace std;
int main() {
    int n; // number of rows store karne ke liye
    cout << "Enter number of rows: "; // user se rows input lena
    cin >> n; // input store karna
    for(int i = 1; i <= n; i++) { // outer loop (rows control karta hai)
        for(int j = 1; j <= i; j++) { // inner loop (columns / stars control karta hai)
            cout << "*" // star print karna
        }
        cout << endl; // next line (next row ke liye)
    }
    return 0; // program end
}
```

7. CONCLUSION

In this lab, we learned about the **for loop** and **nested for loop** in C++. We understood how loops can be used to repeat tasks efficiently. By implementing different programs such as number printing, prime number checking, and pattern generation, we improved our programming logic and problem-solving skills. This lab also helped us understand how nested loops work in real scenarios.